



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2021 - 2022)

JEE(Advanced)
REVIEW TEST
14-03-2021

JEE(Main + Advanced) : NURTURE COURSE (PHASE : TEPS)

PAPER-2

Time : 3 Hours

Maximum Marks : 198

READ THE INSTRUCTIONS CAREFULLY

GENERAL :

1. This sealed booklet is your Question Paper. Do not break the seal till you are told to do so.
2. Use the Optical Response sheet (ORS) provided separately for answering the questions.
3. Blank spaces are provided within this booklet for rough work.
4. Write your name, form number and sign in the space provided on the back cover of this booklet.
5. After breaking the seal of the booklet, verify that the booklet contains **28** pages and that all the **18** questions in each subject and along with the options are legible. If not, contact the invigilator for replacement of the booklet.
6. You are allowed to take away the Question Paper at the end of the examination.

OPTICAL RESPONSE SHEET :

7. The ORS will be collected by the invigilator at the end of the examination.
8. Do not tamper with or mutilate the ORS. **Do not use the ORS for rough work.**
9. Write your name, form number and sign with pen in the space provided for this purpose on the ORS. **Do not write any of these details anywhere else on the ORS.** Darken the appropriate bubble under each digit of your form number.

DARKENING THE BUBBLES ON THE ORS :

10. Use a **BLACK BALL POINT PEN** to darken the bubbles on the ORS.
11. Darken the bubble  **COMPLETELY.**
12. The correct way of darkening a bubble is as : 
13. The ORS is machine-gradable. Ensure that the bubbles are darkened in the correct way.
14. Darken the bubbles **ONLY IF** you are sure of the answer. There is **NO WAY** to erase or "un-darken" a darkened bubble.
15. Take **$g = 10 \text{ m/s}^2$** unless otherwise stated.

DO NOT BREAK THE SEALS WITHOUT BEING INSTRUCTED TO DO SO BY THE INVIGILATOR

Please see the last page of this booklet for rest of the instructions

SOME USEFUL CONSTANTS

Atomic No. : H = 1, B = 5, C = 6, N = 7, O = 8, F = 9, Al = 13, P = 15, S = 16,
Cl = 17, Br = 35, Xe = 54, Ce = 58

Atomic masses : H = 1, Li = 7, B = 11, C = 12, N = 14, O = 16, F = 19, Na = 23, Mg = 24,
Al = 27, P = 31, S = 32, Cl = 35.5, Ca = 40, Fe = 56, Br = 80, I = 127,
Xe = 131, Ba = 137, Ce = 140, Cu = 63.5, Ne = 20, K = 39, Mn = 55

Space for Rough Work

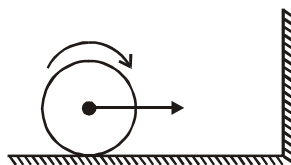
HAVE CONTROL → HAVE PATIENCE → HAVE CONFIDENCE ⇒ 100% SUCCESS

BEWARE OF NEGATIVE MARKING

PART-1 : PHYSICS

SECTION-I : (Maximum Marks: 24)

- This section contains **SIX** questions.
 - Each question has **FOUR** options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s).
 - For each question, choose the correct option(s) to answer the question.
 - Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +4 If only (all) the correct option(s) is (are) chosen.
Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.
Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct options.
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).
Negative Marks : -2 In all other cases.
 - **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.
-
1. A solid sphere is rolling purely on a rough horizontal surface (coefficient of kinetic friction = μ) with speed of center = u . It collides with a smooth vertical wall, the coefficient of restitution being $1/2$. Choose the **CORRECT** option (s) :



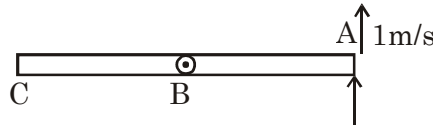
- (A) Speed of COM just after collision is $u/2$
 (B) Friction acts rightwards just after collision
 (C) Angular velocity does not change appreciably during collision
 (D) The speed of center of sphere when it begins pure rolling does not depend on μ .

Space for Rough Work

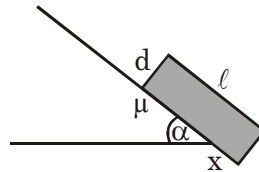
2. In three different containers there are three samples of air at three different temperature values of 300 K, 500 K and 720 K. Allowing heat transfer between the 300 K and the 500 K samples of gas, the common temperature would be 420 K, and allowing heat transfer between the 300 K and the 720 K samples of gas the common temperature would be 600 K.
- (A) The common temperature of the samples of air of initial temperatures of 500 K and 720 K would be 637.5 K
- (B) The ratio of the initial internal energy values of the three samples of gas is 1 : 4 : 3.
- (C) The masses of the three samples of gas are in ratio 2 : 3 : 5.
- (D) The common temperature of the samples of air of initial temperatures of 500 K and 720 K would be 652.5 K
3. A vehicle has metal wheels with $\alpha = 1.2 \times 10^{-5} / ^\circ\text{C}$. It runs at speed of 20 m/s at all the times and covers a distance of 200 km in each run. Its radius is 50 cm at 0°C and if the wheels roll purely then :
- (A) When the temperature increases, the angular velocity of wheels decreases.
- (B) When the temperature increases, the angular velocity of wheels increases.
- (C) At 50°C , it rotates approx by 39.98×10^5 rad to cover 200 km.
- (D) The angular velocity at 50°C is approx. 39.98 rad/s

Space for Rough Work

4. Two alike uniform rods are attached to each other with a pivot joint B at their ends. The rods are at rest on a frictionless horizontal tabletop, lying along a straight line. Suddenly the free end A of one of the rods is hit perpendicularly to the rod such that this endpoint A of the rod begins to move at a speed of 1 m/s.



- (A) The pivot joint B begins to move in the direction opposite to that of A with a speed of $\frac{2}{7} \text{ m/s}$
- (B) The other end C of the other rod begins to move in the same direction as that of A with a speed of $\frac{1}{7} \text{ m/s}$
- (C) The center of mass of the rod which was hit begins to move with a speed of $\frac{5}{7} \text{ m/s}$
- (D) The center of mass of the other rod begins to move with a speed of $\frac{1}{14} \text{ m/s}$
5. Ending at the rim of the table there is a slope of angle of elevation of α , from which a uniform density rectangular block of length ℓ and of height d is sliding down. By what length x does the block move further from the end of the table until it tilts.



- (A) If the friction between the slope and the block is negligible $x = \ell/2$
- (B) If the friction between the slope and the block is negligible $x = (\ell - d \tan \alpha)/2$
- (C) If the coefficient of kinetic friction between the slope and the block is μ ($0 < \mu < \tan \alpha$, and $\mu d < \ell$), $x = \frac{\ell}{2}$
- (D) If the coefficient of kinetic friction between the slope and the block is μ ($0 < \mu < \tan \alpha$, and $\mu d < \ell$), $x = \frac{\ell - \mu d}{2}$

Space for Rough Work

6. The three graphs shown here show the total energy output of four different stars per second plotted against the wavelength that they radiate. All the four stars are behaving like black bodies. They can be assumed to be spherical bodies. Study the graphs below and choose the correct statement(s). Take $b = 3 \times 10^{-3} \text{ mK}$. :-

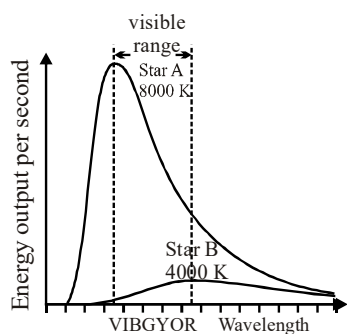


Figure-1

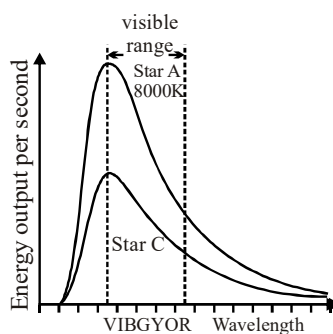


Figure-2

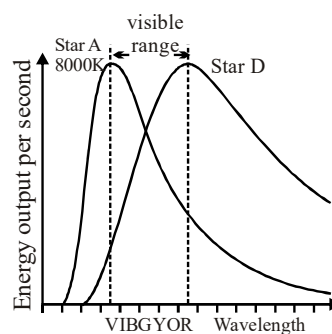


Figure-3

- (A) The peak wavelength of star A and star B respectively are 375 nm and 750 nm
 (B) Star C appears to be blue in colour and has temperature less than 8000 K
 (C) The correct order of the radius of star A, star C and star D are $R_D > R_A > R_C$
 (D) Star D has a temperature more than 8000 K.

Space for Rough Work

SECTION-II : (Maximum Marks: 24)

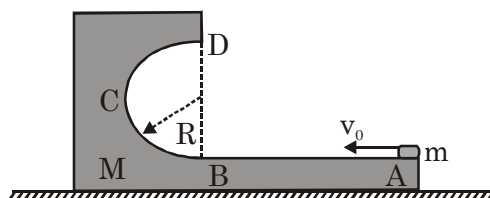
- This section contains **SIX** questions.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct) by darken the corresponding bubbles in the ORS.

For Example : If answer is -77.25, 5.2 then fill the bubbles as follows.

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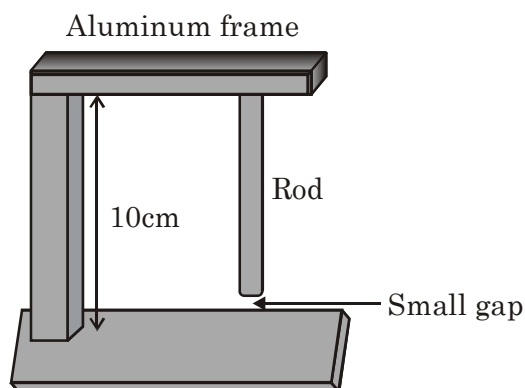
- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +4 If ONLY the correct numerical value is entered as answer.
Zero Marks : 0 In all other cases.

- Figure shows a small block of mass $m = 1\text{ kg}$ which is given a horizontal velocity $v_0 = 10\text{ m/s}$ on the horizontal part of the bigger block of mass $M = 9\text{ kg}$ placed on a horizontal floor. The curved part of the surface shown is semicircular of radius $R = 1\text{ m}$. Find the distance (in m) from point B where the block m lands finally after looping the semicircular part BCD. Neglect friction everywhere. Assume that the horizontal portion AB is long enough. ($g = 10\text{ m/s}^2$)



Space for Rough Work

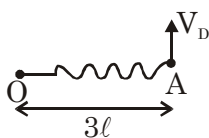
2. At a certain temperature, a rod is hung from an aluminum frame, as the drawing shows. A small gap of 1mm exists between the rod and the floor. The frame and rod are heated uniformly. At what temperature change (in $^{\circ}\text{C}$) will the rod touch the floor, assuming that the rod is made from lead ? The coefficient of thermal expansion of aluminum is $23 \times 10^{-6}/^{\circ}\text{C}$ and for lead it is $29 \times 10^{-6}/^{\circ}\text{C}$. The height of the frame is 10 cm.



3. A heating element is built into a boiler of 10 liters. The element has such a small power that it is unable to heat the water to its boiling point. After heating the water totally, during the first minute after ceasing the heating the temperature of the water decreases by 1°C . What is the power rating (in W) of the heating element if the water equivalent of the calorimeter is 3kg?

Space for Rough Work

4. The specific heat capacity of metals at very low temperatures is approximately proportional to the absolute temperature $c = \alpha.T$, where the proportionality constant α is characteristic of the material). In a very well insulated chamber of a cryogenic laboratory, two pieces of different metals of different mass are placed such that they came into contact. The initial temperature of one of them (denoted by A) is 1.0 K, whilst that of the other (B) is 3.0 K, and the final common temperature is 2.0 K. What will the final common temperature (in K) be if the initial temperature values of the metals are $T_A = 1.5K$ and $T_B = 2.5 K$?
5. A particle of mass m is connected with a string of length 5ℓ . One end of the string is fixed at O and other end is kept at a distance 3ℓ in a vertical plane at A as shown in figure. OA is horizontal initially. Particle is given a velocity of $\sqrt{8\ell g}$ in vertically upward direction. Tension in the string at lowest point is x mg. Find value of x :



6. After the warm days, a sharp frost hit a lake and its surface got covered with ice. One day after the cold snap; the ice thickness was $d_1 = 2$ cm. Some ship builders want to take weights on the opposite shore of the lake. But for security reasons ice thickness of at least $d_2 = 10$ cm is needed. How many days after moment of frost is it possible to transport the goods, if the temperature of surrounding does not change.

Space for Rough Work

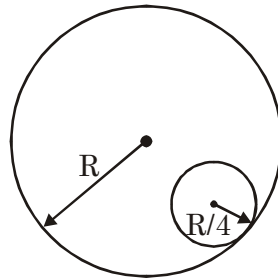
SECTION-III : (Maximum Marks : 18)

- This section contains **SIX** questions.
- The answer to each question is a **SINGLE DIGIT INTEGER** ranging from 0 to 9, both inclusive.
- For each question, darken the bubble corresponding to the correct integer in the ORS.
- For each question, marks will be awarded in one of the following categories :
Full Marks : +3 If only the bubble corresponding to the correct answer is darkened.
Zero Marks : 0 If none of the bubbles is darkened.
Negative Marks : -1 In all other cases

1. The inside part of a circular-base vertical tube of height 1 m and of inner area $\frac{2}{9} \text{ cm}^2$ is filled with ice of temperature 0°C . The wall of the tube is slippery and is made of some insulating material. The tube is open at its top and closed at its bottom. We start heating the bottom of the tube at a rate of 3360 W. (Density of ice = 0.9 gm/cc and density of water is 1 gm/cc & $4.2 \text{ J} = 1 \text{ cal}$) At what constant speed (in cm/s) will the top of the ice-cylinder move downwards?
2. A (trolley + child) of total mass 200 kg is moving with a uniform speed of 36 km/h on a frictionless track. The child of mass 40 kg starts running on the trolley from one end to the other (10 m away) with a speed of 10 ms^{-1} relative to the trolley in the direction of the trolley's motion and jumps out of the trolley with the same relative velocity. What is the final speed (in m/s) of the trolley ?

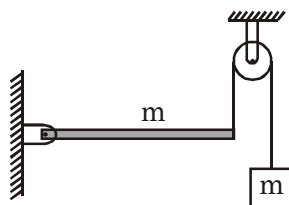
Space for Rough Work

3. The blood plays an important role in removing heat from the body by bringing this energy directly to the surface where it can radiate away. Nevertheless, this heat must still travel through the skin before it can radiate away. Assume that the blood is brought to the bottom layer of skin at 37.0°C and that the outer surface of the skin is at 30.0°C . Skin varies in thickness from 0.50 mm to a few millimeters on the palms and soles, so assume an average thickness of 0.75 mm. A 165-lb, 6-ft-tall person has a surface area of about 2.0 m^2 and loses heat at a net rate of 75 W while resting. On the basis of our assumptions, what is the thermal conductivity (in 10^{-3} W/mK) of this person's skin ?
4. A ring of mass m and radius $(R/4)$ rolls inside a fixed hoop of radius R such that at the highest point of its trajectory the normal reaction becomes zero. The friction on the hoop is sufficient to ensure pure rolling. The angular velocity of the ring at the topmost point is $\omega = 2\sqrt{\frac{\eta g}{R}}$, then find the value of η .

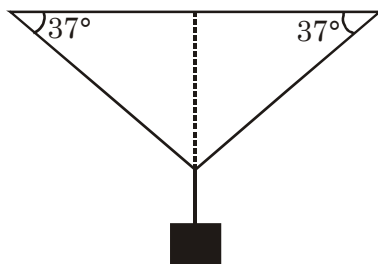


Space for Rough Work

5. A rod and a block are same mass. The rod is hinged at the other end. Initially rod is in horizontal position. When system is released from this position, the acceleration of tip of the rod is $\frac{3g}{K}$. Find the value of K.



6. A block is hanged by means of two identical wires having cross section area A (1 mm^2) as shown in the diagram. If temperature is lowered by ΔT (10°C), find the mass (in kg) to be added to hanging mass such that junction remains at initial position. Given that co-efficient of linear expansion $\alpha = 1 \times 10^{-5} / ^\circ\text{C}$ and Young's modulus $Y = 5 \times 10^{11} \text{ N/m}^2$ for the wire.



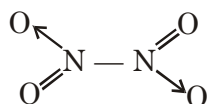
Space for Rough Work

PART-2 : CHEMISTRY

SECTION-I : (Maximum Marks: 24)

- This section contains **SIX** questions.
- Each question has **FOUR** options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s).
- For each question, choose the correct option(s) to answer the question.
- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +4 If only (all) the correct option(s) is (are) chosen.
Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.
Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct options.
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).
Negative Marks : -2 In all other cases.
- **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

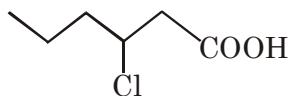
1. Correct order(s) of boiling point is/are :
 (A) $\text{CCl}_4 > \text{SiCl}_4$ (B) $\text{F}_2 < \text{Cl}_2$ (C) $\text{H}_2 > \text{He}$ (D) $\text{NH}_3 < \text{SbH}_3$
2. If following structure lies in XY plane, then select correct option/s.



- (A) All N-O bond lengths are equal
- (B) Nodal plane of all $\text{N} \overset{\pi}{-} \text{O}$ bonds lie in XY plane
- (C) Bond order of all N-O bonds is equal to 1.33
- (D) N-N bond has no nodal plane

Space for Rough Work

3. Which of the following IUPAC name is/are incorrect?



(A) 3-Chlorohexan-1-oic acid

(B) 4-Chlorohexan-1-oic acid

(C) 3-Chloropentane-1-oic acid

(D) 3-Chloropentan-5-carboxylic acid

4. Correct IUPAC name is/are ?

(A) 2-Methylpentane

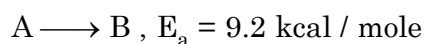
(B) 1-Chlorohexane

(C) 3-Methylcyclohex-1-ene

(D) 2-Methylbutan-1-al

Space for Rough Work

5. For first order reaction



At 500 K rate constant for reaction is $6.93 \times 10^{-4} \text{ s}^{-1}$. Which of the following option(s) is/are correct ? (Use : $\ln X = 2.3 \log X$, $\ln 2 = 0.693$, $R = 2 \text{ Cal/ mol / K}$)

(A) Value of rate constant for reaction at 1000 K is $6.93 \times 10^{-2} \text{ s}^{-1}$

(B) In 20 seconds , 75 % decomposition of A will occur at 1000 K

(C) For 99.9 % completion at 500 K, 10^4 seconds are required.

(D) In 30 seconds, 87.5 % decomposition of A will occur at 1000 K

6. A metal exists in two allotropic forms. α -form is BCC and β -form is FCC. The distance between nearest neighbours is same in both forms. Which of the following is/are **INCORRECT** information(s) related with these two forms of the metal ?

(A) The distances between next nearest neighbours is same in both.

(B) Percentage void space in α -form is greater than that in β -form.

(C) The diameter of largest sphere which may be fitted in the crystals of metal, without disturbing the distance between nearest neighbours, is same in both forms.

(D) The cell constant is higher for the unit cell of β -form.

Space for Rough Work

SECTION-II : (Maximum Marks: 24)

- This section contains **SIX** questions.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct) by darkening the corresponding bubbles in the ORS.

For Example : If answer is -77.25, 5.2 then fill the bubbles as follows.

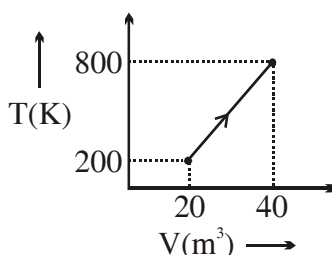
+	●				
●	●	0	0	0	0
1	1	1	1	1	1
2	2	2	2	●	2
3	3	3	3	3	3
4	4	4	4	4	4
5	5	5	5	5	●
6	6	6	6	6	6
7	7	●	●	7	7
8	8	8	8	8	8
9	9	9	9	9	9

				-	
●	●	●	0	0	●
1	1	1	1	1	1
2	2	2	2	●	2
3	3	3	3	3	3
4	4	4	4	4	4
5	5	5	●	5	5
6	6	6	6	6	6
7	7	7	7	7	7
8	8	8	8	8	8
9	9	9	9	9	9

- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +4 If **ONLY** the correct numerical value is entered as answer.
Zero Marks : 0 In all other cases.

1. Calculate magnitude of work done in calorie by one mole of an ideal gas subjected to the process as shown in figure.

[Given: $R = 2 \text{ Cal/ mol / K}$, $\ln 2 = 0.7$]



2. When photon of energy 4.25 eV strike the surface of a metal A, the ejected photoelectrons have maximum kinetic energy, T_A (in eV) and the de-broglie wavelength λ_A . The maximum kinetic energy of the photoelectron emitted from another metal B by photons of energy 4.7V is $T_B = (T_A - 1.5) \text{ eV}$. If the de broglie wavelength of these photoelectrons is, $\lambda_B = 2\lambda_A$ then work function of metal B is.

Space for Rough Work

3. If $\text{Si}_x\text{O}_y^{6-}$ is basic formulae unit of amphibole silicate then find the value of $\frac{y}{x}$.

4. Find the value of expression $\frac{(x+z)}{(y+w)}$ for following oxides.

B_2O_3 , OCl_2 , CO , PbO , Cr_2O_3 , MnO , HgO , Mn_2O_7 , Cs_2O

where

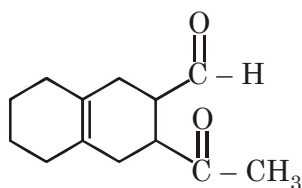
x = total number of acidic oxides

y = total number of basic oxides

z = total number of amphoteric oxides

w = total number of neutral oxides

5. Identify D.U. (Degree of unsaturation) of following compound?



6. Number of σ -bond's in following compound

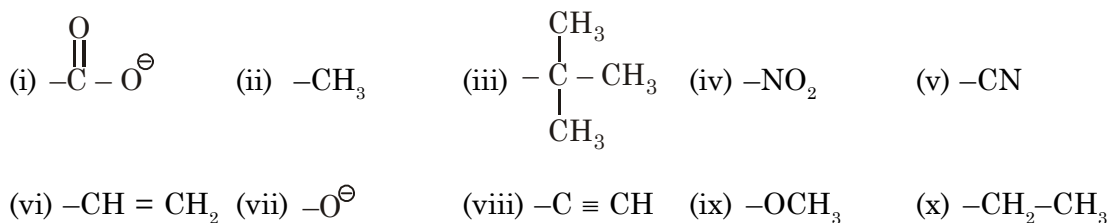


Space for Rough Work

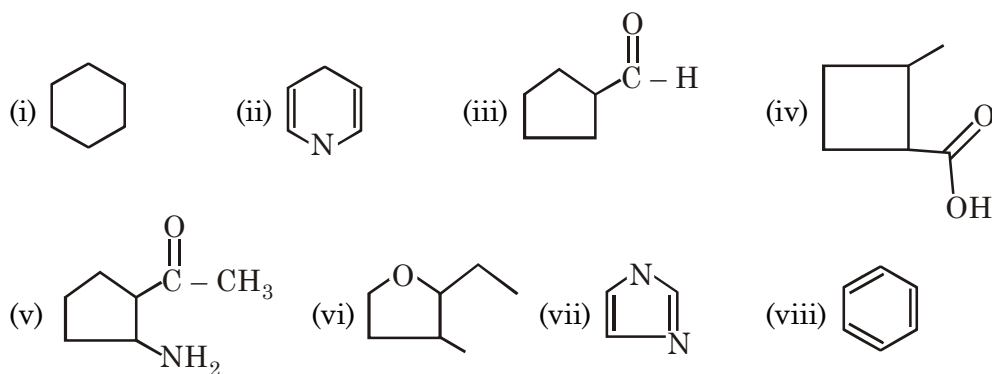
SECTION-III : (Maximum Marks : 18)

- This section contains **SIX** questions.
- The answer to each question is a **SINGLE DIGIT INTEGER** ranging from 0 to 9, both inclusive.
- For each question, darken the bubble corresponding to the correct integer in the ORS.
- For each question, marks will be awarded in one of the following categories :
Full Marks : +3 If only the bubble corresponding to the correct answer is darkened.
Zero Marks : 0 If none of the bubbles is darkened.
Negative Marks : -1 In all other cases

1. Find out total number of +I groups in the given groups ?



2. Number of compounds which is/are homocyclic ?



Space for Rough Work

3. Find out total number of polar molecular species having more than 50% p-character in their respective bond pair.
 NH_2^+ , H_2S , ClO_2F_2^- , XeF_3^+ , NOCl , SiH_4 , POF_3 , FCIO_2 , ICl_4^-
4. Find the number of acids from the following in which X-H bond is present. (X is P, S or B)
 H_3PO_2 , $\text{H}_4\text{P}_2\text{O}_7$, $\text{H}_4\text{P}_2\text{O}_5$, $\text{H}_2\text{S}_2\text{O}_6$, H_3PO_3 , H_3BO_3 , H_2SO_4
5. A radioactive isotope ${}_Z\text{A}^{\text{M}}(t_{1/2} = 10 \text{ days})$ decays to give ${}_{(Z-6)}\text{B}^{(\text{M}-12)}$ stable atom. If m grams of 'A' is taken and kept in a sealed tube and $5.6 \times 'x'$ liters of He gas at 1 atm and 0°C is accumulated in 20 days, then the value of x is.
6. Enthalpy for the reaction, $\text{A}^+(\text{aq.}) + \text{B}^-(\text{aq}) \rightarrow \text{AB}(\text{s})$ is -20 kcal . Magnitude of enthalpy of formation of $\text{A}^+(\text{aq})$ and $\text{B}^-(\text{aq})$ are in the ratio 4 : 5. Formation of $\text{A}^+(\text{aq})$ is an endothermic process whereas formation of $\text{B}^-(\text{aq})$ is an exothermic process enthalpy of formation of $\text{AB}(\text{s})$ is $-21.5 \text{ kcal mol}^{-1}$. Calculate the enthalpy of formation of $\text{A}^+(\text{aq})$ in Kcal mol^{-1} .

Space for Rough Work

PART-3 : MATHEMATICS

SECTION-I : (Maximum Marks: 24)

- This section contains **SIX** questions.
- Each question has **FOUR** options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s).
- For each question, choose the correct option(s) to answer the question.
- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +4 If only (all) the correct option(s) is (are) chosen.
Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.
Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct options.
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).
Negative Marks : -2 In all other cases.
- **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

1. If sides AB, BC and CA of a triangle ABC are represented by $x + 2 = 0$, $3x + y = 0$ and $x + 3y + 2 = 0$ respectively, then identify the correct statement -

(A) $\sum \tan A = \frac{4}{3}$

(B) $\prod \tan A = -\frac{4}{3}$

(C) $\sum \tan A \tan B = -\frac{41}{9}$

(D) $\sin^2(A+B) + \cos^2 C = \frac{5}{4}$

2. If $A_1, A_2, \dots, A_n$ are n -distinct odd natural numbers not divisible by any prime greater than 5, then $\frac{1}{A_1} + \frac{1}{A_2} + \frac{1}{A_3} + \dots + \frac{1}{A_n}$ can never exceed -

(A) 1

(B) 2

(C) 3

(D) 4

Space for Rough Work

3. Let a, b, c be any three positive digits, then six digit number of the form 'abcabc' is divisible by-
(A) 77 (B) 11 (C) 13 (D) 17
4. Consider a system of equations : $kx + \lambda y = k + \lambda$ & $\lambda x + ky = 1$, ($k, \lambda \in \mathbb{R}$) then identify the incorrect statements.
(A) Value of $k + \lambda$ for which system is inconsistent cannot be 1
(B) Value of $\frac{k}{\lambda}$ is always 1 if the system is inconsistent
(C) Value of $k + \lambda$ for which the system is inconsistent can be equal to 1 or 0
(D) If two points (x_1, y_1) & (x_2, y_2) satisfy the given system then $k + \lambda$ is either 0 or 1

Space for Rough Work

5. If $\begin{vmatrix} \frac{a}{b} + \frac{a}{c} & \frac{b}{c} + \frac{b}{a} & 1 \\ \frac{b}{a} + \frac{b}{c} & \frac{c}{a} + \frac{c}{b} & 1 \\ \frac{c}{a} + \frac{c}{b} & \frac{a}{b} + \frac{a}{c} & 1 \end{vmatrix} = 0$, where $a, b, c \in \mathbb{R}^+$, then which of the following is/are necessarily true -

(A) $\frac{1}{a} + \frac{1}{b} + \frac{1}{c} = 0$

(B) $a^2 + b^2 + c^2 = ab + bc + ac$

(C) $a = b = c$

(D) $\frac{1}{a} + \frac{1}{b} + \frac{1}{c} = 1$

6. If two sides of a regular hexagon have the equation $2x + y - 1 = 0$ & $4x + 2y - 5 = 0$, then -

(A) its area is $\frac{9\sqrt{3}}{40}$

(B) distance of centre of hexagon from any of its sides is $\frac{3}{4\sqrt{5}}$

(C) the line parallel to given sides & passing through the centre is $8x + 4y - 7 = 0$

(D) perimeter of hexagon is $\sqrt{\frac{27}{5}}$

Space for Rough Work

SECTION-II : (Maximum Marks: 24)

- This section contains **SIX** questions.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct) by darkening the corresponding bubbles in the ORS.

For Example : If answer is -77.25, 5.2 then fill the bubbles as follows.

+		-	
●	●	●	●
①	①	①	①
②	②	②	②
③	③	③	③
④	④	④	④
⑤	⑤	⑤	⑤
⑥	⑥	⑥	⑥
⑦	⑦	⑦	⑦
⑧	⑧	⑧	⑧
⑨	⑨	⑨	⑨

- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +4 If **ONLY** the correct numerical value is entered as answer.
Zero Marks : 0 In all other cases.

- Sum of all real values of λ , so that the equation $|x - \lambda^2| + |x - 4\lambda| = |\lambda|$ has infinite number of solutions for x is
- Consider ΔABC with vertices $A(3, 2)$, $B(5, 0)$ and $C(0, -1)$. Vertex A is reflected in side BC , vertex B is reflected in side AC and vertex C is reflected in AB . Let the reflected points be respectively P , Q and R . Then the area of ΔPQR will be

Space for Rough Work

3. If x be a real number such that $x^3 + 4x = 8$, then the value of the expression $x^7 + 64x^2$ is -
4. The medians of a $\triangle ABC$ are 9 cm, 12 cm and 15 cm respectively. Then the area of the triangle is -
5. The product of two of the four roots of $x^4 + kx^2 + 590x = 20x^3 + 1992$ is 24, then k is -
6. If the line $y = \sqrt{3}x$ cuts the curve $x^4 + ax^2y + bxy + cx + dy + 6 = 0$ at A, B, C and D, then value of $OA \cdot OB \cdot OC \cdot OD$ is, (where O is origin) -
-

Space for Rough Work

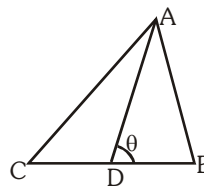
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Negative Marks : -1 In all other cases
-

1. Number of straight lines passing through the fixed point (3, 2) & cutting positive integral intercepts on coordinate axes is
 2. Let triangle ABC be the isosceles triangle and $\angle A = 100^\circ$, if $\sin A + \sin B + \sin C = x$ and $\cos A + \cos B + \cos C = y$, then $(x^2 + y^2)$ is
 3. Number of all possible solutions of the equation $\cot\left(\frac{\pi}{3\sqrt{3}}\sin 2x\right) = \sqrt{3}$ in the interval $[0, 2\pi]$ is -
-

Space for Rough Work

4. In the adjacent figure, if $AC = 5$, $AB = 4$, $CD = DB$ & area of $\triangle ABC$ is 8sq. units, then value of $32 \cot \theta$ is



5. Let $P(x)$ be a cubic polynomial with zeros α, β, γ . If $\frac{P\left(\frac{1}{2}\right) + P\left(-\frac{1}{2}\right)}{P(0)} = 100$, then the value of

$$\frac{\left(\frac{1}{\beta\gamma} + \frac{1}{\gamma\alpha} + \frac{1}{\alpha\beta}\right)}{49}$$
 is equal to

6. The equations of two adjacent sides of a rhombus formed in first quadrant are represented by $7x^2 - 8xy + y^2 = 0$ then slope of its longer diagonal is -

Space for Rough Work

Space for Rough Work

QUESTION PAPER FORMAT AND MARKING SCHEME :

16. The question paper has three parts : Physics, Chemistry and Mathematics.

17. Each part has three sections as detailed in the following table.

Section	Que. Type	No. of Que.	Category-wise Marks for Each Question				Maximum Marks of the section
			Full Marks	Partial Marks	Zero Marks	Negative Marks	
I	One or more correct option(s)	6	+4 If only the bubble(s) corresponding to all the correct option(s) is(are) darkened	+1 For darkening a bubble corresponding to each correct option, provided NO incorrect option darkened	0 If none of the bubbles is darkened	-2 In all other cases	24
II	Numerical Value Type (Up to second decimal place)	6	+4 If only the bubble corresponding to correct answer is darkened	—	0 In all other cases	—	24
III	Single digit Integer (0-9)	6	+3 If only the bubble corresponding to correct answer is darkened	—	0 If none of the bubbles is darkened	-1 In all other cases	18

NAME OF THE CANDIDATE

FORM NO.

I have read all the instructions and shall abide by them.

Signature of the Candidate

I have verified the identity, name and Form number of the candidate, and that question paper and ORS codes are the same.

Signature of the Invigilator